



Pop-Up Coach

User manual | One-page summary

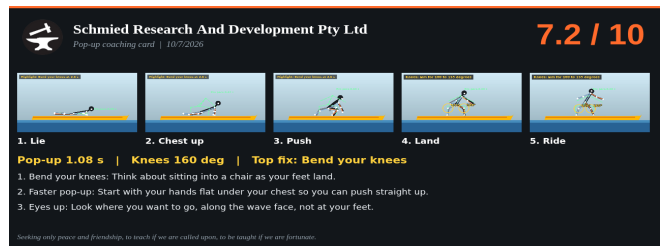
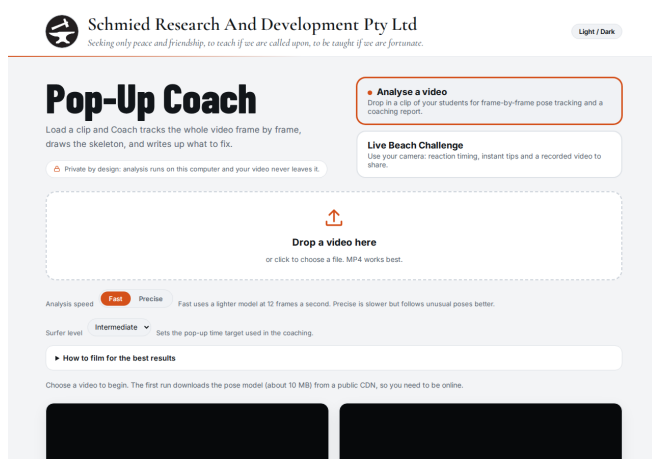
Schmied Research And Development Pty Ltd

Seeking only peace and friendship, to teach if we are called upon, to be taught if we are fortunate.

Pop-Up Coach turns a video of a surfer into a frame-by-frame coaching report. It tracks the body, draws a skeleton, races a professional-style ghost, scores the pop-up out of 10 and tells the surfer what to fix, with tips, drills and a practice plan. Everything is analysed on your own device. Videos and camera images are never uploaded.

Get started in six steps

- 1 **Open the app** in Chrome, Edge or Safari from its web address (or the saved file) and install it from the browser. The first run needs internet.
- 2 **Choose a mode:** Analyse a video, or Live Beach Challenge.
- 3 **Set the surfer level** (Beginner, Intermediate or Advanced) and the analysis speed (Fast or Precise), then drop in a side-on clip with the whole body in frame.
- 4 **Press Play.** Watch the skeleton, the green pro ghost and the yellow highlights that show where to improve.
- 5 **Read the Coach's report:** pro-style score, the six measures behind it, tips, drills and a practice plan.
- 6 **Save and share.** Save the session under the student's name, then share the coaching card, the video or a WhatsApp summary.



Left: the home screen. Right: a shareable coaching card. Screenshots use a demonstration clip.

Analysis

Pose tracking and skeleton, filming-quality check, stance detection, squad report for every surfer in frame.

Progress

Student profiles, a trend chart that marks uploaded videos and Live Beach Challenges, leaderboard, before and after comparison, backup, and suggested recognition of returning students.

Coaching

Pro-style score out of 10, tips, drills, practice plan, yellow highlights and a pro ghost that races you at 0.6 s.

Sharing

Coaching card (PNG), annotated video (MP4 or WebM, beside the original or a skeleton-only view), full report as a PDF, WhatsApp summary, printed report.

Remember

Film side-on from 3 to 5 metres, whole body in frame, steady, good light. **10 / 10** means a professional-style pop-up. Pop-up time targets: Beginner 1.3 s, Intermediate 1.0 s, Advanced 0.8 s. **Live camera** needs an https address; on a phone you can also record with the camera app and load the clip. The Live Beach Challenge speaks every command and writes its own Coach's report.

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1 Getting started and installing

Pop-Up Coach runs entirely in your web browser. There is nothing to buy and no account to create. It works on Windows, Mac, Chromebook, Android and iPhone.

What you need

- A current browser: **Chrome** or **Edge** on a computer, **Chrome** on Android, **Safari** on iPhone and iPad.
- Internet for the **first run** only. The app downloads the pose engine and model (about 10 MB). The folder version then keeps them on the device so it can work offline.
- An **https** address if you want the live camera. Browsers only allow cameras on secure pages.

Installing it

- **Windows, Mac, Chromebook:** open the address in Chrome or Edge and press **Install app** in the page header, or use the install icon in the address bar.
- **Android:** open the address in Chrome and press **Install app** (or the browser menu, then Install app).
- **iPhone and iPad:** open the address in Safari, tap **Share**, then **Add to Home Screen**.
- **Saved file (email or WhatsApp copy):** double-click it on a computer and it opens in your browser. iPhones cannot run saved HTML files.

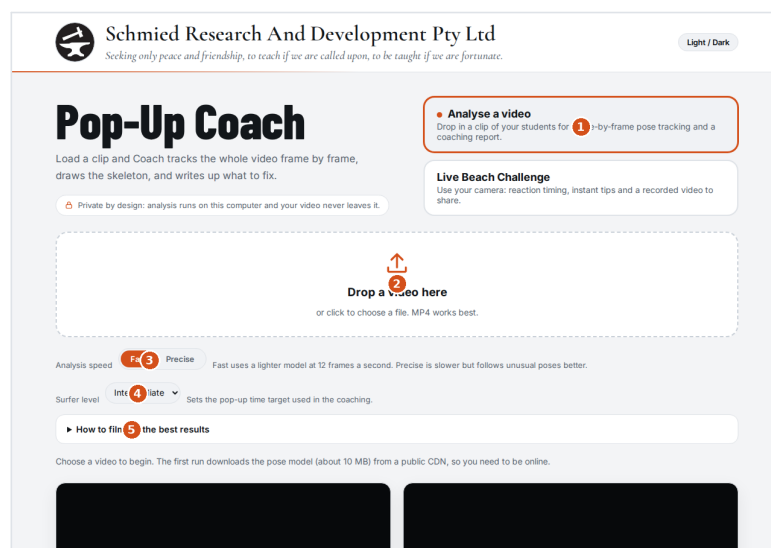


Figure 1. The home screen. (1) Mode tiles. (2) Drop zone: drag a video here or click to choose one. (3) Analysis speed. (4) Surfer level. (5) Filming tips.

Keeping up to date

The footer shows the version number, for example **Version 2026.10.08a**. If you host the app yourself, upload the new **index.html** and **sw.js** over the old files, then reload. If you installed the app, remove it and install again to refresh it.

2 Analysing a video

Filming for the best results

- Film from the **side** (side-on), at chest height, 3 to 5 metres away.
- Keep the **whole body** in frame from head to toes, and the board.
- Hold the camera steady, in landscape, in good light, at 30 frames a second or more.
- Film one surfer at a time if you can, and a few pop-ups in a row. Short clips (under 45 seconds) give the sharpest results.

Loading and analysing

- 1 Choose **Analyse a video**.
- 2 Set the **Surfer level**. It sets the pop-up time target used in the coaching: Beginner 1.3 s, Intermediate 1.0 s, Advanced 0.8 s.
- 3 Set the **Analysis speed**. **Fast** uses a lighter model at 12 frames a second. **Precise** uses the full model at 15 frames a second and follows unusual poses better.
- 4 Drop the video on the drop zone, or click it to choose a file. MP4 (H.264) works best.
- 5 Wait for the progress bar. The status line shows the percentage and the seconds left. Your device does the work.

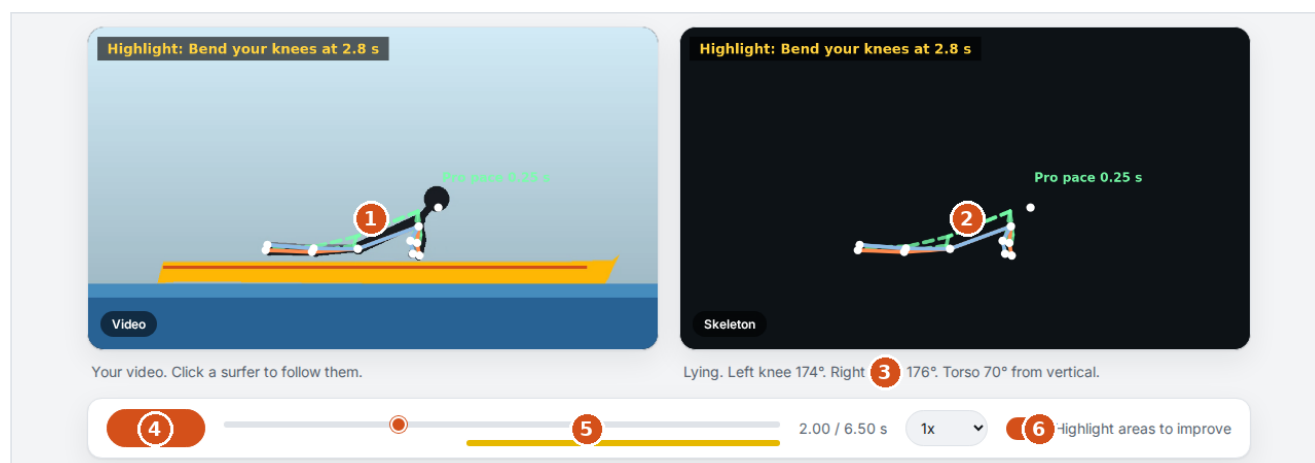


Figure 2. The analysis view. (1) Your video. (2) The skeleton panel. (3) Live readout: phase, knee angles and torso angle. (4) Play button. (5) Scrubber, with yellow marks for the moments to study. (6) Highlight switch. The speed menu is beside it.

Controls

Play / Pause	Space bar or the Play button	Step one frame	Left and right arrow keys
Speed	Menu beside the scrubber: 0.25x, 0.5x, 1x	Follow another surfer	Click them in the video
Jump to phase	Lie, Chest up, Push, Land, Ride buttons under the controls	Light or dark	Light / Dark button in the header

3 Skeleton, pro ghost and highlights

The **skeleton** follows the surfer in the video and is also drawn on its own dark panel. Orange lines are the left side and blue lines the right side. If several people are in the shot, small dots mark the others. Click a dot or a person to follow them instead.

The pro pop-up ghost

Switch on **Pro pop-up ghost** in the Share and squad card. A green dashed figure pops up from the moment the surfer leaves the board, at pro pace (0.6 seconds), lands both feet together in a low crouch with knees near 100 degrees, and keeps its eyes up. The label shows the pro pace clock. Compare the surfer with the ghost frame by frame.

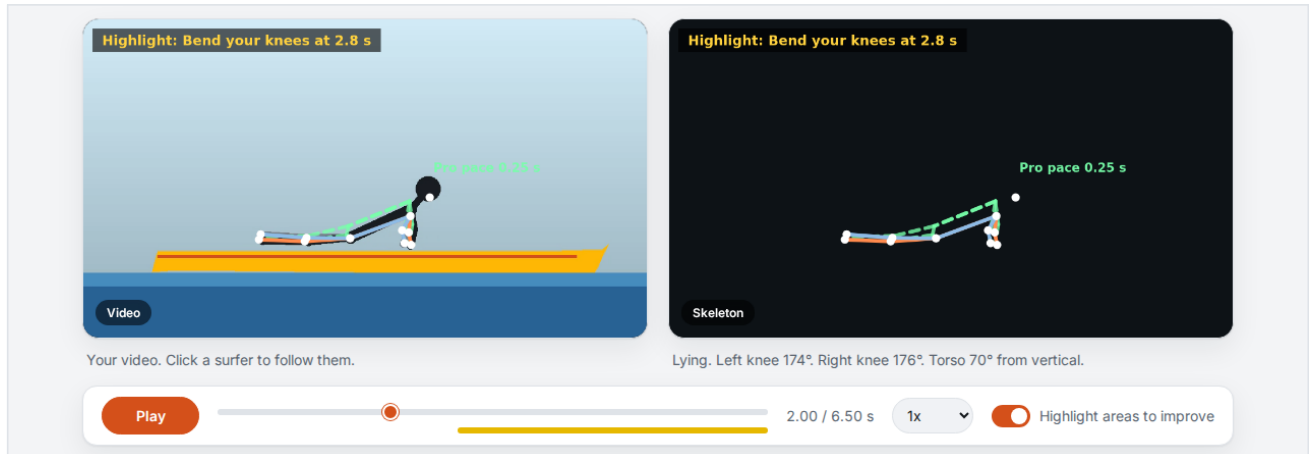


Figure 3. The green ghost racing the surfer. At 2.0 seconds the ghost is already part way through its pop-up while the surfer is still pushing up.

Two ghost modes: **Race the pro pop-up** (the ghost runs at pro pace from where the surfer starts) and **Ideal pose for my phase** (the ghost shows the ideal pose for whichever phase the surfer is in).

Yellow highlights

Each of the top three improvements has a yellow mark that appears on the video and the skeleton while the problem is happening. Knee bend rings the knees with their angle. Stance draws a line between the feet. Chest position shows the torso against a vertical guide. Head position rings the head. Steadiness traces the hip path. Pop-up speed boxes the surfer and runs a timer. A label on the video names the highlight and when it happens.

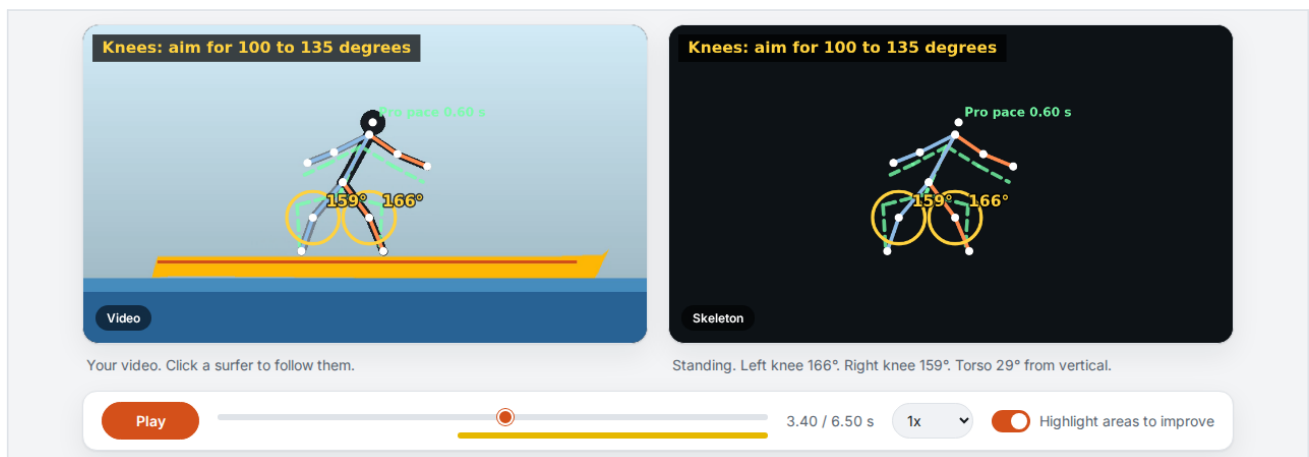


Figure 4. A highlight: both knees are ringed with their angles, and the banner says what to aim for (here, 100 to 135 degrees).

Press **Show on video** on any improvement card to switch the highlight and jump to it. Turn the **Highlight areas to improve** switch off to hide the marks.

4 Reading the Coach's report

The report appears under the video as soon as the analysis finishes.

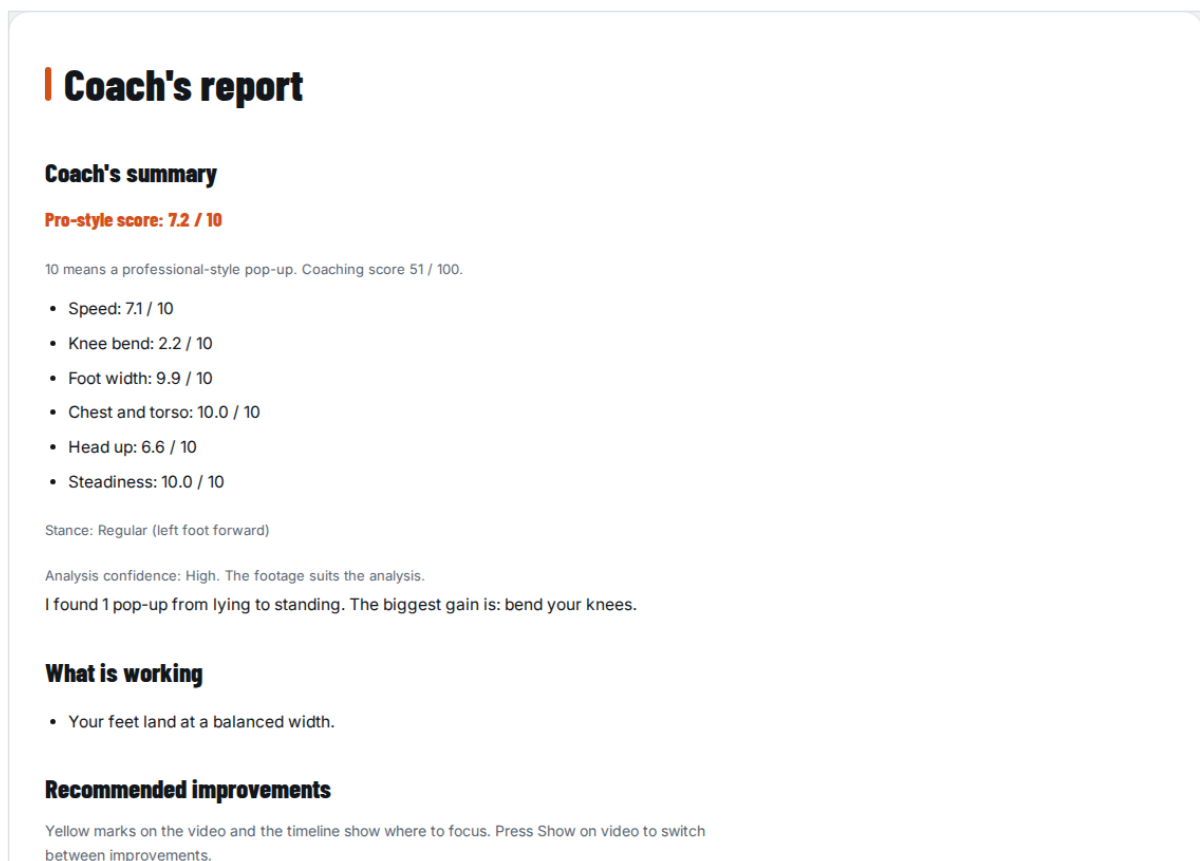


Figure 5. The top of the report: pro-style score, the six measures behind it, and what is working.

The pro-style score

A score from 0 to 10. **10 means a professional-style pop-up.** It combines six measures, each scored out of 10: speed (30 percent), knee bend (20), foot width (15), chest and torso (15), head up (10) and steadiness (10). The older **coaching score out of 100** is kept for the trend chart and leaderboard.

Recommended improvements

The top three fixes, most important first (Figure 6). Each card explains what the video shows, gives two or three coaching **tips**, and a **drill** to practise on the beach. **Show on video** jumps to the moment and switches the highlight.

Your practice plan

A warm-up, one session for each of the top three fixes (three sets of ten, 30 seconds rest), and a final step to film again from the same angle and compare scores.

Analysis confidence

The report rates the footage High, Medium or Low. It checks that the surfer was tracked throughout, is big enough in the frame, is well lit, and was filmed side-on. If confidence is not High, it says how to film better.

1

Bend your knees

After you stand, your legs are nearly straight (about 160 degrees at the knee). Stiff legs cannot absorb the bumps.

Tips

- Think about sitting into a chair as your feet land.
- Keep your weight over the balls of your feet.
- Let your knees track over your toes, like a skier.

Drill: Hold a low surf stance for 20 seconds, ten times. Knees over toes, chest over the front thigh.

Show on video

2

Faster pop-up

Your pop-up averages 1.08 s from lying to standing. A good target for your level is 1.0 s or less, because a slow pop-up costs you the line.

Tips

- Start with your hands flat under your chest so you can push straight up.
- Pop in one motion: hands, then both feet together, with no knee stop in between.
- Keep your eyes on the horizon, not your feet.

Drill: Beach reps: lie flat and pop up in one count. Ten reps, three sets. Film it and aim to get under one second.

Show on video

3

Eyes up

Your head drops toward your shoulders once you are standing, which pulls your weight forward.

Tips

- Look where you want to go, along the wave face, not at your feet.
- Lift your chin slightly and keep your neck long.

Drill: Pick a cone or a spot on the horizon and pop up looking at it. Ten reps.

Show on video

Your practice plan

1

Warm up for five minutes: arm circles, hip hinges and five slow beach pop-ups.

2

Session 1, bend your knees: Hold a low surf stance for 20 seconds, ten times. Knees over toes, chest over the front thigh. Do three sets of ten, resting 30 seconds between sets.

3

Session 2, faster pop-up: Beach reps: lie flat and pop up in one count. Ten reps, three sets. Film it and aim to get under one second. Do three sets of ten, resting 30 seconds between sets.

4

Session 3, eyes up: Pick a cone or a spot on the horizon and pop up looking at it. Ten reps. Do three sets of ten, resting 30 seconds between sets.

5

Film one more clip from the same angle and compare your score and pop-up time with today.

Figure 6. Improvement cards with tips and drills, the practice plan, and the pop-ups found.

5 Sharing and exporting

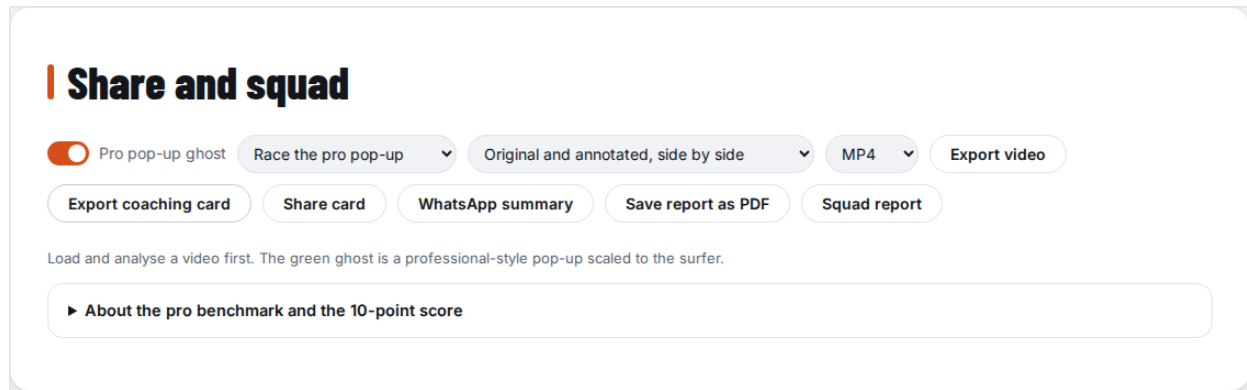


Figure 7. The Share and squad card.

Coaching card

Export coaching card saves a PNG with the five phases (Lie, Chest up, Push, Land, Ride), the score and the top three tips, with your logo and motto. **Share card** opens the phone or computer share sheet so you can send it straight to WhatsApp, email or messages. If sharing files is not supported, the card is saved instead.

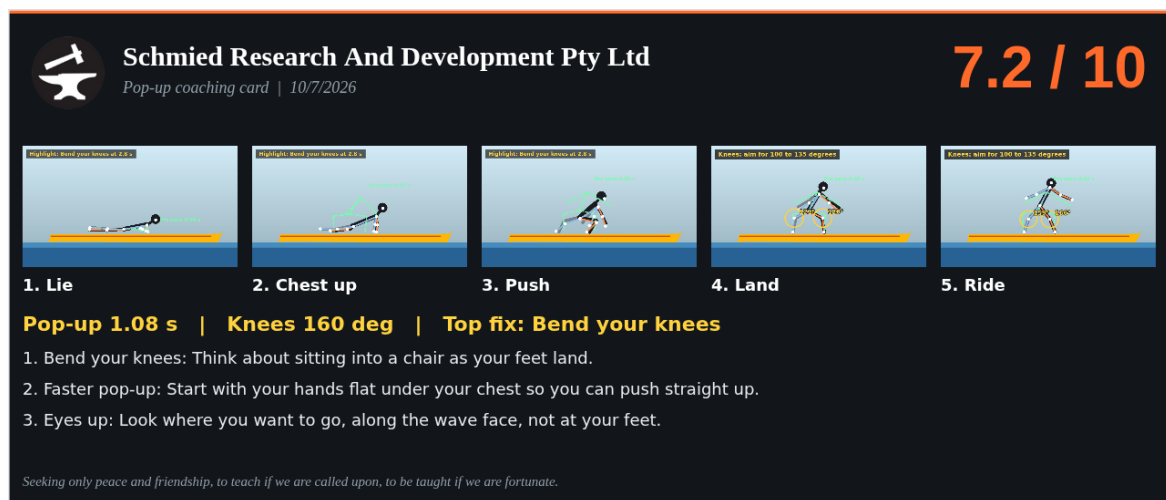


Figure 8. A coaching card, ready to send.

Video export

Choose the layout and format, then press **Export video**.

- **Layout.** **Original and annotated, side by side** puts the plain video beside the annotated one. **Annotated and skeleton only, side by side** puts the annotated video beside the dark skeleton panel. **Annotated only** exports just the annotated video.
- The company name and logo sit in their own strip across the top of every saved video, so they never cover the comments.
- **MP4 or WebM.** MP4 plays almost everywhere. If your browser cannot record a format, that option is greyed out and marked not supported.
- The export plays the video through in real time while recording, so a 30-second clip takes about 30 seconds.

Full report as a PDF

Press **Save report as PDF** to download the whole Coach's report as a document: the pro-style score and the six measures, what is working, the top three improvements with tips and drills, the practice plan and the pop-ups found. It also includes **screen grabs of the annotated video and the skeleton only, side by side**, at each phase of the pop-up. The PDF is built on your device, so it works offline.

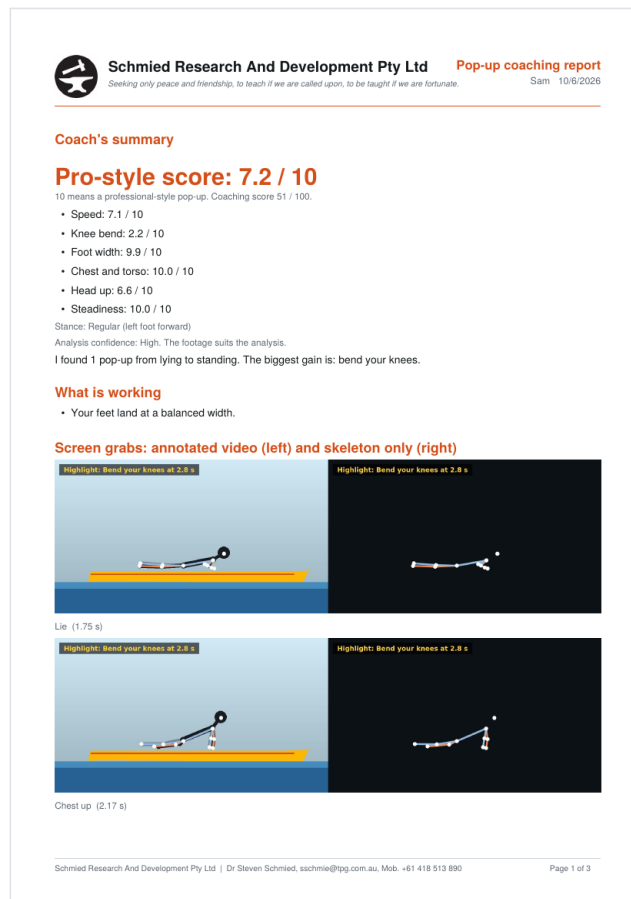


Figure 9. The first page of a saved PDF report.

Other ways to share

- WhatsApp summary** opens WhatsApp with the report text ready to send.
- Print report** (in the Students and progress card) makes a branded page. Choose Save as PDF in the print window to keep a copy.

6 Students, progress and recognition

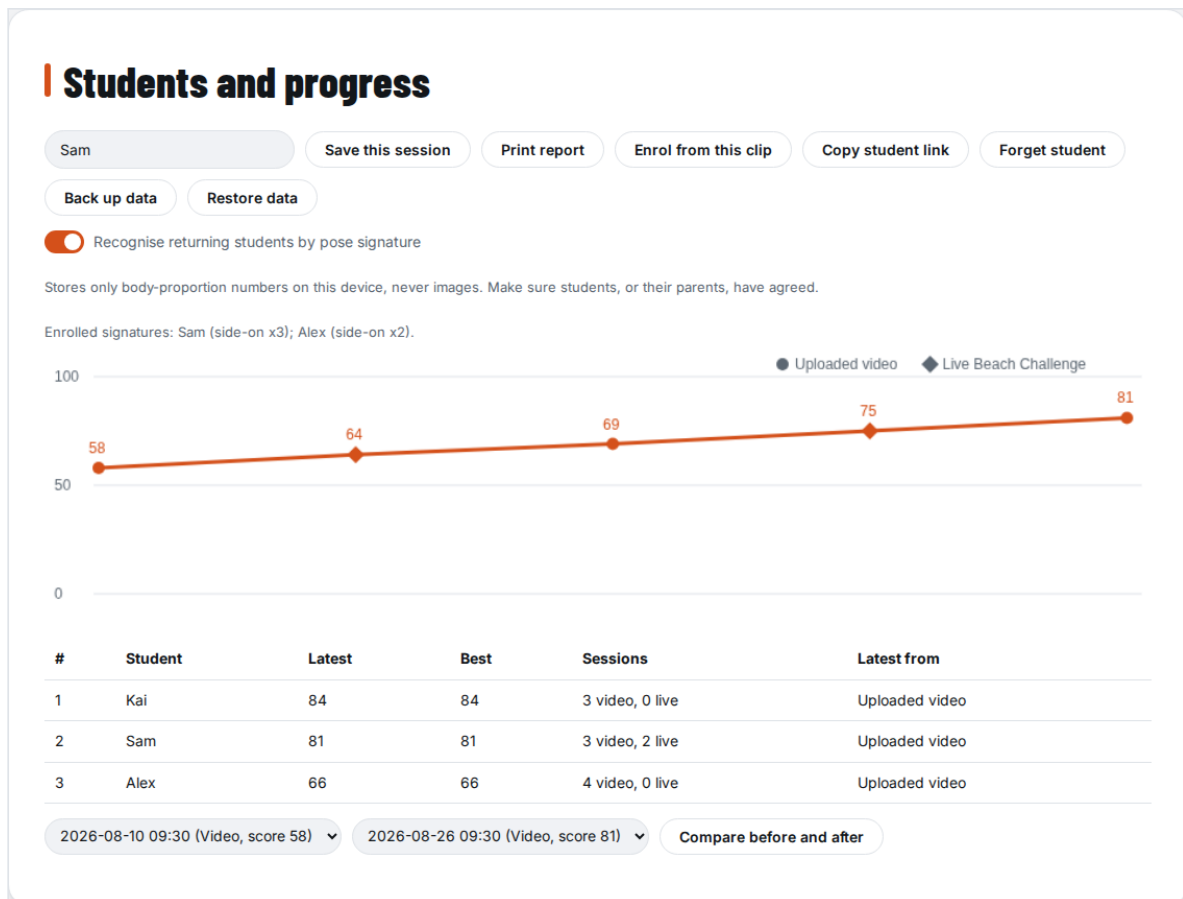


Figure 10. Students and progress: saved sessions, a score trend, the leaderboard and the compare tool.

Saving sessions

Type the student's name and press **Save this session**. Sessions are stored on **this device**. The trend chart shows the coaching score over time. The leaderboard ranks every student by their latest score. To compare, pick a before and an after session and press **Compare before and after**.

Where each score came from

Every saved session is marked as coming from an **Uploaded video** or a **Live Beach Challenge**. The leaderboard shows how many sessions came from each and where the latest one came from. The trend chart uses a circle for an uploaded video and a diamond for a Live Beach Challenge, with a key above the chart. The before and after menus label each session Video or Live. Sessions saved before this feature are treated as uploaded videos.

Welcome back and student links

When saved students exist, a banner says **Welcome back** and shows quick-pick buttons. **Copy student link** makes an address such as `.../?student=Sam` that opens Sam's profile on a device that already holds their sessions.

Recognising returning students

With **Recognise returning students by pose signature** on, the app can suggest who a surfer is from body proportions (limb lengths compared with the torso). It always asks you to confirm: *This looks like Sam. Is it them?* If two students look alike it offers both. Only the same camera angle is compared. Press **Enrol from this clip** to add a signature without saving a session, and film each student side-on two or three times for the best results. Signatures are only numbers. No images are stored.

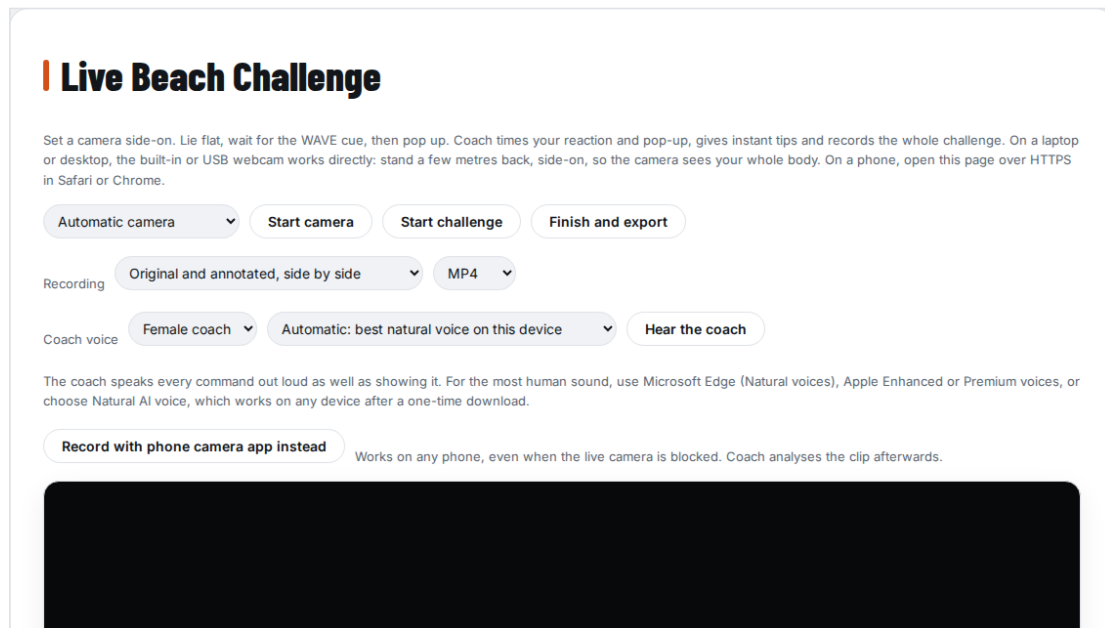
Privacy and consent

Make sure students, or their parents or guardians, have agreed before you use recognition, especially for under-18s. Use **Forget student** to delete a student's sessions and signature. Use **Back up data** and **Restore data** to move profiles to another device, because clearing browser data removes them.

Squad report

Squad report finds every surfer in the busiest frame, scores each, names them from saved signatures (each name used once) and lets you save each to a profile.

7 Live Beach Challenge



The screenshot shows the 'Live Beach Challenge' control panel. At the top, there's a title 'Live Beach Challenge' with a vertical bar icon. Below it, a paragraph explains the setup: 'Set a camera side-on. Lie flat, wait for the WAVE cue, then pop up. Coach times your reaction and pop-up, gives instant tips and records the whole challenge. On a laptop or desktop, the built-in or USB webcam works directly: stand a few metres back, side-on, so the camera sees your whole body. On a phone, open this page over HTTPS in Safari or Chrome.' The interface includes several controls: a camera selection dropdown (currently 'Automatic camera'), 'Start camera', 'Start challenge', and 'Finish and export' buttons. A 'Recording' section has a layout dropdown ('Original and annotated, side by side') and a format dropdown ('MP4'). A 'Coach voice' section has a voice selection dropdown ('Female coach'), a quality dropdown ('Automatic: best natural voice on this device'), and a 'Hear the coach' button. A note mentions using Microsoft Edge or Apple Enhanced/Premium voices for better sound. At the bottom, there's a button 'Record with phone camera app instead' and a note 'Works on any phone, even when the live camera is blocked. Coach analyses the clip afterwards.' A large black rectangular area at the bottom represents the video feed.

Figure 11. The Live Beach Challenge controls, including the coach voice.

Choose **Live Beach Challenge** at the top of the page. Coach times your reaction and pop-up, **speaks every command out loud**, gives an instant tip after every rep, records the whole challenge, and writes a **Coach's report** when you finish.

- 1 Pick a camera: **Automatic**, **Laptop or front webcam**, or **Rear camera**. After the camera starts, every camera found is listed by name.
- 2 Pick the **coach voice**: **Female coach**, **Male coach** or **Voice off**, and the voice quality (see The spoken coach below). Press **Hear the coach** to test it.
- 3 Press **Start camera** and allow access when the browser asks. Set the camera side-on, a few metres away, so it sees the whole body.
- 4 Choose the recording **layout** and **format** (MP4 or WebM), then press **Start challenge**.
- 5 Lie flat on your stomach. The coach says **Lie flat**, then **Hold**, then **Wave! Pop up!** Jumping early gets **Too early**.
- 6 After each rep the coach says your reaction time, your pop-up time and one tip, and the same appears on screen. Repeat as often as you like.
- 7 Press **Finish and export** for the recorded video, the Coach's report, a coaching card from your best rep, and a button to save that rep to a profile.

Commands and the saved video

The commands sit in a panel at the **left** of the picture. In every saved video the company name and logo are in their own strip across the top, so they can never cover the commands or the comments. The commands appear in all three layouts: original and annotated, annotated and skeleton only, and annotated only.

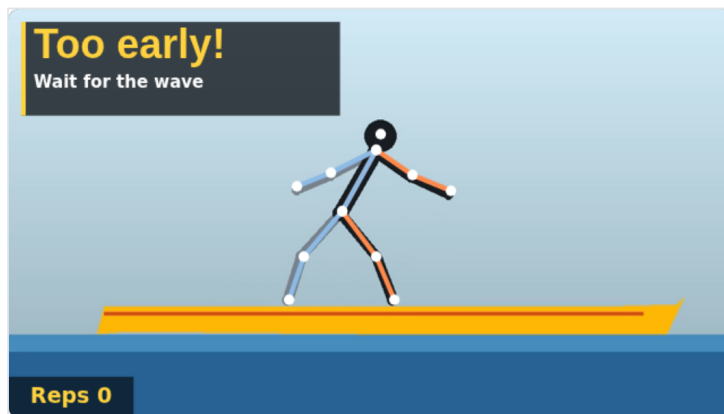


Figure 12. The live view. The command panel is at the left; the caption at the bottom counts reps and the best score.

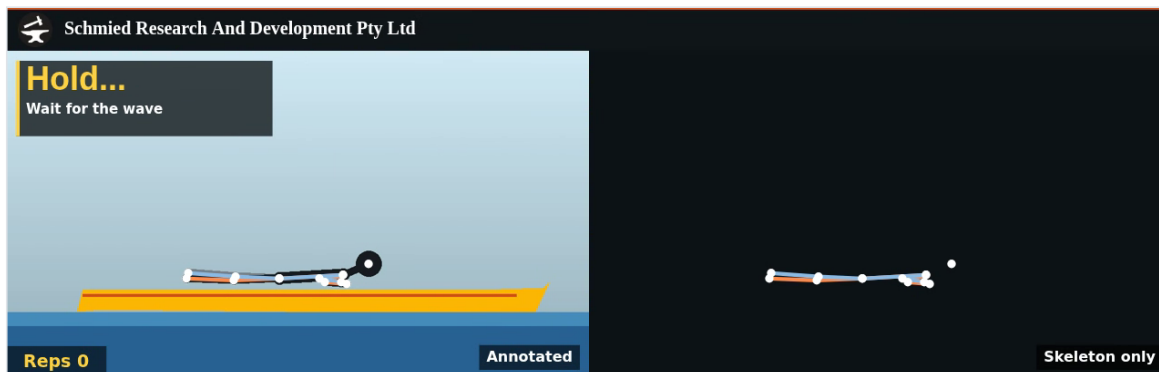


Figure 13. A frame from a saved video: logo strip on top, the commands at the left, the annotated video beside the skeleton only.

The spoken coach

The coach says **Lie flat**, **Hold** and **Wave! Pop up!**, tells you when you are **too early** or miss a wave, and after each rep gives your reaction time, your pop-up time and one tip. The wording changes a little each time, so it sounds less like a machine. Use the second menu to choose how the coach sounds:

- **Automatic.** The app ranks the English voices on your device and picks the most natural one that matches Female or Male coach. It prefers Microsoft Edge **Natural** voices, Apple **Enhanced** and **Premium** voices and Google voices over older built-in ones, and it prefers Australian English.
- **Natural AI voice.** A human-sounding voice that runs on your own device and works the same on any computer or phone. The first time you choose it, it downloads about 90 MB, then it is kept for next time. Until it is ready, the device voice is used.
- **A voice from the list.** Every English voice found on the device is listed, with a star beside the most natural ones.

For the most human sound on a device voice, use **Microsoft Edge** on Windows (the Natural voices are free), or add an Enhanced or Premium voice on an iPhone or Mac (Settings, Accessibility, Spoken Content, Voices). Voices come from your device, so they differ between devices. The first sound plays after you press **Start challenge**.

Saving the video even with no pop-ups

Press **Finish and export** at any time. Even when no pop-up was completed, the recorded video is saved and can be downloaded, and the report explains what to try next time.

The Coach's report for the challenge

After every challenge the app writes a report: the pro-style score (speed, knee bend, chest and head position), your average reaction to the wave, what is working, up to three improvements with tips and drills, a practice plan and a list of reps. Press **Save Coach's report as PDF** to keep it, with screen grabs of your best rep showing the annotated video and the skeleton only, side by side. **Save best rep to profile** records the session as coming from a Live Beach Challenge.

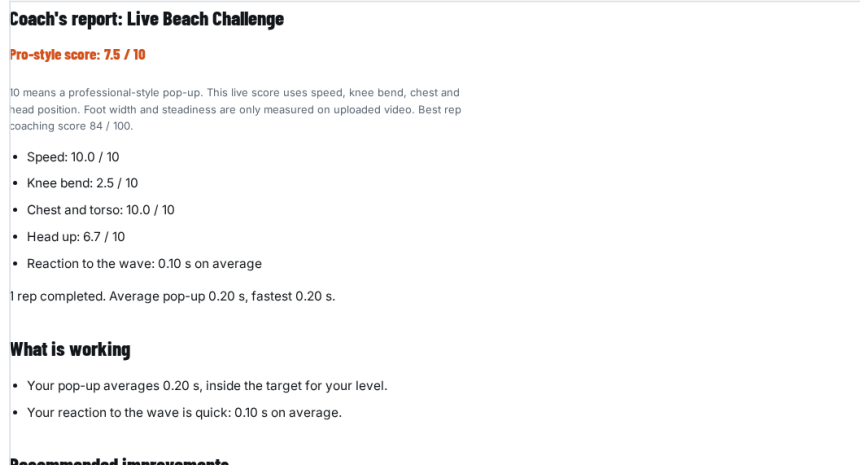


Figure 14. The top of a Live Beach Challenge report.

Good to know

The camera only works on https addresses (and on localhost). On a phone, if the live camera is blocked, use **Record with phone camera app instead**, then analyse the clip as a normal video. When you finish, the app may suggest a profile from the pose signature in the live video.

8 Using it on a phone

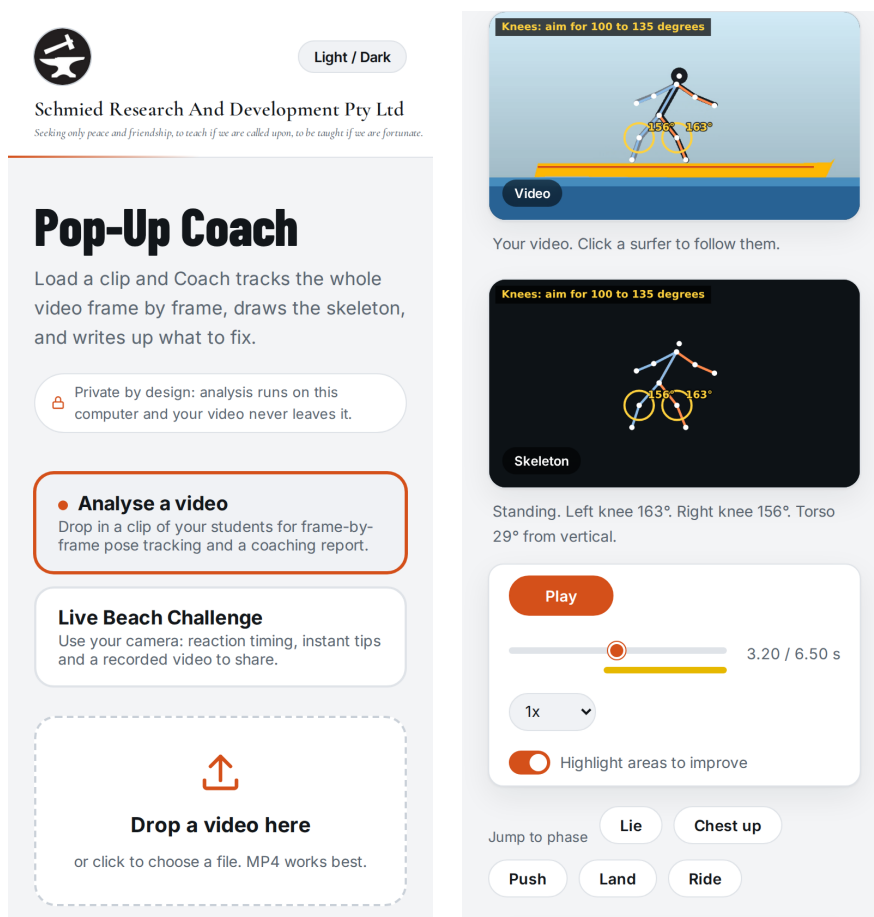


Figure 15. The home screen and the analysis view on a phone.

- Open the app from its **web address in Safari or Chrome**. Links opened inside another app (WhatsApp, Gmail, social apps) often block the camera and file features. Use the menu and choose Open in browser.
- iPhones cannot run a saved HTML file from Mail, WhatsApp or Files. Use the web address instead.

Layout. The page stacks into one column on a phone. The mode tiles, drop zone, video, skeleton, report and cards follow one below another.

Install. Android Chrome: Install app. iPhone Safari: Share, then Add to Home Screen.

Film. Prop the phone side-on in landscape. A tripod or a friend helps.

Share. Share card opens the phone share sheet, so you can send to WhatsApp in two taps.

- Videos recorded on an iPhone may be HEVC. On the iPhone itself Safari plays them. To use them on other devices set **Settings, Camera, Formats, Most Compatible**.

9 Troubleshooting

"This video uses HEVC (H.265)" or "This browser cannot play this video"	The browser cannot play the file. Convert it to H.264 with HandBrake (free, preset Fast 1080p30), or open the app in Safari on a Mac or iPhone, or in Edge on Windows with the HEVC Video Extensions. The Details line under the message shows the codecs found in the file.
The live camera does not start	Camera access needs an https address (or localhost). Allow the permission when asked, close other apps using the camera, and try the other camera in the menu. On a phone use Record with phone camera app instead .
The coach sounds robotic	Open the second voice menu and choose Natural AI voice (a one-time download of about 90 MB), or use Microsoft Edge on Windows for its Natural voices, or pick a starred voice from the list. If the natural voice cannot load, the app says so and falls back to the best device voice.
The coach does not speak	Check the device volume and that Voice off is not selected. Press Hear the coach to test. Voices depend on the device and browser; if none is installed the app cannot speak. The first sound only plays after you press Start challenge .
"Could not load the pose model"	The first run needs internet to fetch the pose engine. Connect, reload and try again.
Analysis is slow	Use Fast , trim the clip to the pop-ups, close other tabs, and keep the tab in front while it runs. The app uses the graphics card automatically when it can.
"I could not find a full pop-up"	The surfer must be seen lying and then standing, with the whole body in frame. Film side-on, or click the right surfer in the video. The app still measures stance when someone is standing.
A student is suggested wrongly	Press Someone else . Recognition compares body proportions only and cannot tell similar builds apart. Enrol each student from two or three side-on clips.
Sessions disappeared	Profiles live in the browser on that device. Clearing browser data removes them. Use Back up data regularly.
The page looks old, or a fix is missing	Check the version in the footer. Reload, reinstall the app, or upload the newest files if you host it.
Offline use	The folder version saves the pose files after the first run and then works offline. The single-file version needs a connection to open the page.

10 About the benchmark, privacy and limits

Where the pro benchmark comes from

Pop-Up Coach was not built from measurements of professional surfers. The benchmark comes from published research on experienced surfers and from coaching accounts of the one-motion "sliding" pop-up used on tour.

- **Speed:** in-water time to pop-up, from the chest leaving the board to front-foot contact, averaged about 0.6 s in one study of experienced surfers (Parsonage and colleagues, 2020).
- **Laboratory pop-up:** a simulated pop-up took about 1.2 s overall, with knees near 99 degrees and elbows near 110 degrees at peak push (Sensors, 2021).
- **Scoring ranges:** the ranges used for the 10-point score (for example knees 95 to 125 degrees) are the author's coaching estimates, not published measurements of professionals.

Limits you should know

- Everything is measured from a **single 2D camera view**. Angles and distances are approximate, and side-on footage works best.
- The app measures speed to the moment the torso is upright, which is slightly later than foot contact, so scores can read a little low.
- Pose tracking can be wrong when limbs are hidden, the surfer is small, or the light is poor. Check the confidence line.
- Recognition by pose signature is a suggestion that needs confirmation. It cannot separate similar builds and should never be the only way you identify a student.
- This is a coaching aid. It is not medical advice and does not replace a qualified coach.

Privacy

Videos, camera images and pose data are processed on your device. Saved sessions and signatures stay in your browser. Nothing is sent to a server by the app. The only network use is downloading the pose engine and model on the first run, and fonts. A shared link such as the student link contains only a name.

Contact

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